

High Throughput Screening of Chromatographic Phases for Rapid Process Development

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The necessity of being “first on the market” paired with the need to cut production costs highlights the need for strategies and techniques allowing a fast process development while at the same time covering a high number of potential process parameters in order to reach an optimal process sequence. This can only be met by the use of high throughput screening (HTS) techniques. In this paper HTS applications presently applied in bioprocess development are reviewed, their use in chromatographic process parameter evaluation is discussed and concepts for the rational design of HTS experiments are presented. A rational approach for chromatographic parameter screening comprehends an initial screening procedure for ligand selection followed by the incorporation of specific adsorbent features using HTS pipetting systems finalized by a fine tuning step using packed mini-columns. The analytical part of these screening techniques clearly sets the bottleneck of these procedures with a trade off between analytical information and analysis time. Finally, intelligent HTS strategies should in addition include efficient data processing paired with modern tools for design of experiments or model based process design.

1 Introduction

The production of proteins as biopharmaceuticals has become one of the most profitable fields in the pharmaceutical industry. In 2003, the global market was valued at about US\$ 41 billion comprising 120 products with the prospect of rising up to US\$ 100 billion until the end of the decade [1]. These rising global sales are admittedly going parallel with increasing costs for research and development (R&D) due to a permanently rising number of legal regulations. In 2002 the R&D costs for bringing a new drug to market have risen up to US\$ 802 million [2]. In a typical biopharmaceutical process such as the production of monoclonal antibodies, the costs can be split into one third cell culture, one third purification and one third support [3]. Processing of monoclonal antibodies, however, is based on a so-called platform strategy putting “Protein A” affinity chromatography in the heart of the overall purification process. This reduces costs otherwise needed for process development and led to an effective overall process. On the other hand more complex purification tasks might shift the contribution of downstream processing to the overall cost to considerably higher percentages. Consequently downstream processing is considered an “unresolved bottleneck” [4] allowing significant improvements in process economics. To achieve this goal existing strategies have to be streamlined to combine the “first to market” requirements with the ability to reach overall optimized processes. For this task HTS techniques seem to be the perfect candidate.

HTS is a technology that had its breakthrough in the late 1990s in the analytical sector and for lead discovery [5]. This breakthrough only became possible due to the development of new robotic based liquid handling techniques, labware

concepts such as the 96/384/1536 well formats and rapid data analysis. In its extreme case, this approach led to a daily throughput of more than 100,000 assays per day described as ultra high throughput screening (uHTS). To date these numbers have only been reached in the analytical sector.

In the field of bioprocess development HTS has been mainly used as a screening tool for strain development and cultivation strategies [6]. These approaches cover the search for strains expressing enzymes with a higher activity or enantioselectivity [7–9], the optimization of bioconversion processes by screening for high product formation rates [10, 11] or substrate screening experiments for the development of fermentation processes [12]. Being only excerpts from a wide variety of miniaturization or automation attempts in bioprocess development these examples clearly show the high potential of HTS techniques.

The use of HTS techniques for protein purification was mainly influenced by the need to cope with increasing size and number of combinatorial libraries and the purification of recombinant proteins carrying an affinity tag [13]. These applications, however, focused mainly on the number of targets handled using the same protocol rather than a screening of different parameters leading to an optimized purification process. Today, combinatorial libraries can be purified with a sample throughput of about 250 samples per instrument and day [14].

One of the first applications targeting the development of purification processes based on HTS screening arose from the use of crystallization as a routine technique for the sample preparation prior to x-ray analysis. Following this strategy Klyushnichenko et al. [15] have worked on the crystallization of therapeutic proteins reaching a production scale of 150 L.

Few initial attempts to introduce “low throughput” screening of chromatographic resins for a given task can already be found in literature [16–18]. First steps using HTS platforms or experimental design techniques were only reported recently [19–21].

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A successful implementation of HTS in chromatographic process development, however, can only be realized by combining the data generation with the “brain work” in an experimental cycle, which covers a variety of experimental techniques, analytics, data handling and experimental design strategies.

2 Chromatographic Process Development

Chromatographic processes belong to the most widely used systems applied in industrial purification processes in the biotechnological sector. The outcome of a chromatographic separation is naturally influenced by a variety of parameters characterizing the overall process. Systemizing these parameters and their influence on the overall process performance, one might differentiate between three levels. On a nano-scale level the interaction between the spacer-ligand complex and the target molecule determine the actual molecular interaction. Zooming out to a meso-scale level the adsorbent structure or pore design, the chemical backbone and of course the ‘ability’ of the investigated molecules to deal with the respective adsorbent features (diffusivities, adsorption and desorption) determine the chromatographic separation of the overall process to a major extend. Finally, the sum of the single separation processes is shown on a macro scale in the performance of a packed bed including hydrodynamic parameters such as convection and dispersion within the column. Depending on the scope of the actual screening/process development study, boundary conditions must thus be adjusted to the respective “scale” of information required.

2.1 Ligand Screening

In the core of chromatographic separations lies the interaction between the ligand and the target molecule. The latest development investigating these interactions as an approach to aid process development is called “surface-enhanced laser desorption-ionization retentate chromatographyTM mass spectrometry” [22]. The approach is based on the assumption that the overall governing factor in chromatographic separation processes lies in the actual protein-ligand interaction. Planar surfaces in microchip format are functionalized with the respective chemical groups present in commercially available resins. Protein solutions are bound to the microchip followed by a step-wise elution. Proteins remaining on the chip after the elution step are analyzed via mass spectrometry. Doing so, the best suited functional group and optimum binding conditions can be found as well as the ionic strength best suited for predominant binding of the target protein and the lowest ionic strength needed for elution of the protein. Combining rapid sample analysis with a low detection limit this technique, however, neglects the significant impact of three dimensional pore structure of

chromatographic adsorbents and differences arising from interactions with the backbone. To date only low resolution applications such as capture steps have been optimized using this approach.

2.2 Resin Screening

Taking the influence of pore structure, adsorbent design – perfusive pores, tentacle structures or gel in the shell approaches – and of the adsorbent backbone into account, it becomes necessary to widen system boundaries to studies investigating protein-adsorber interactions [23]. This will not only create a direct link to real process performance in a packed column, but will also add resin capacity, kinetic performance, resolution and yield to the investigated parameter field.

The simplest approach retrieving the relevant information lies in the use of finite bath experiments. High throughput screening of commercially available resins on the basis of finite bath studies uses the power of modern lab robotic platforms [19,20]. These stations combine the ability to handle solutions and suspensions on a low volume basis, offer features such as solid-liquid separation by vacuum filtration or centrifugation and use analysis based on UV, fluorescence or luminescence. Samples are normally handled in multi-well plate formats, among which the 96-well plate format seems most adequate for automated screening experiments combining a high number of wells (experiments) per plate and a sufficient working volume (< 2 mL).

2.2.1 Plate Handling

Due to the high surface to volume ratio evaporation in microtiter plates is a considerable source of error within HTS applications. Evaporation depends on well position on the plate and liquid level within the well. Further more when increasing the distance to the unsaturated ambient air the driving force for evaporation decreases. Thus, decreasing liquid level within the well decreases loss of liquids noticeably. In a study using a volume of 100 μ L in a 300 μ L standard microtiter plate (MTP) up to 2.5 μ L/h liquid evaporated at room temperature (unpublished data). In different studies it could be shown that moving from the center to the edge of the plate, the overall evaporation might increase by up to 10 % [24,25]. For long time applications, such as mixing and incubation at elevated temperatures, usage of a lid or better a sealing tape prevents evaporation most effectively.

Representative adsorption/desorption processes require complete mixing of the protein-resin suspensions. Generally four different methods might be applied for mixing in wells based on microtiter plate formats: (1) magnetic stirrer, (2) end-over-end shaker, (3) displacement and (4) orbital shaker. Given the rather fragile matrix of certain adsorbent structures, magnetic stirring leads to a high ratio of shredded

adsorbent particles. The application of traditionally used end-over-end shaker exhibits good mixing features but is difficult to implement on commercially available pipetting stations. Displacement techniques express a potential for automation but are hampered by the same problem of solid-solid shear experienced with magnetic stirring. Moreover any additional surface within a well introduces a potential for non-specific binding. Finally, orbital shaker combine a high potential for automation with a low level of shear, which makes them presently the method of choice. In order to cope with the small dimension and the requirement of optimal mixing conditions several studies deal with appropriate concepts developing so-called “in phase” shaking conditions [26,27].

A simple test for “in-phase” shaking conditions can be performed by measuring protein uptake curves in 96-well plates under different conditions. With insufficient mixing protein uptake is hindered, as resin particles settle in the bottom of the wells being thus inaccessible to proteins in solution. Fig. 1 shows such a study where rpm values below 1200 rpm led to insufficient mixing of a 500 μ L solution (containing 8 μ L agarose resin) in a 1.0 mL round-well microtiter plate. Below 1200 rpm protein uptake does not increase over time. Visual inspection showed in this case that the resin particles have settled to the bottom of the wells shortly after addition of the protein solution to the resin. Above 1200 rpm resins remained well distributed in the liquid, providing full protein access into the resin particles for adsorption.

In order to provide a known amount of adsorbent particles per well prior to chromatographic parameter generation two different methods might be applied: a) slurry aliquotation

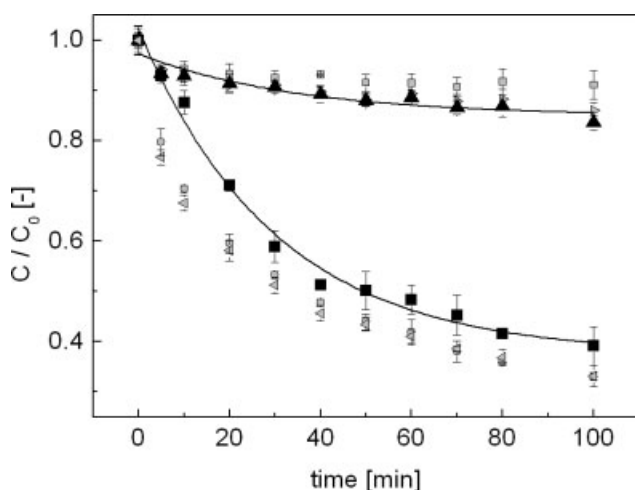


Figure 1. Influence of shaking frequency (□ 900 rpm, ◇ 1000 rpm, ▲ 1100 rpm, ■ 1200 rpm, ○ 1300 rpm, < 1400 rpm) on mixing behaviour in 96-well plate. Decrease of Bovine Serum Albumin (BSA) concentration (c) in supernatant is plotted against incubation time. Below 1200 rpm protein adsorption to Toyopearl® SP-550C hardly takes place. Visual inspection exhibited settled resin at bottom of well. For sufficient shaking in microtiter plates, a threshold rpm has to be exceeded. Data points for 1100 rpm and 1200 rpm are described with an exponential decay (line). Error bars result from quadruple experiments.

b) resin plaques. The method of slurry aliquotation is based on the preparation of adsorbent slurries with known concentration followed by distribution via automated pipetting [19,20]. This technique exhibits a number of drawbacks. The generation of known resin slurries comprises several manual error-prone steps and is difficult to automate. Resin slurries are defined by weight per volume and thus the respective resin densities are needed to relate to volumetric parameters measured in packed bed columns. Finally, pipetting of slurries expresses a high potential for experimental errors. In contrast, the use of resin plaques of defined volume prepared prior to an experimental study has shown to express a superior performance and eliminates the above mentioned drawbacks [28]. Plaques are generated using a grid in 96-well plate format with holes of a defined volume. The holes are then filled with the respective resins by using vacuum [28] or centrifugal force [29].

2.2.2 Parameter Generation

The organization of experiments for parameter generation can be based on two general strategies. The generated parameters may either be used within the framework of model based process development [30,31] (see section on Data Analysis and Experimental Design) or serve as a “self-contained” matrix of experimental data, building up a parameter platform, which covers the most relevant parameters (adsorbent type, interaction mode, buffer composition etc.) for a given task [20].

For a first assessment regarding the suitability of an adsorbent for a given purification task various pieces of information are needed. Primarily *product capacity*, *interaction strength* and *uptake kinetics* for a given combination of fluid condition and used solid phase give a first evidence of suitable adsorbent types. A comparison to the respective data of contaminating molecules leads to first estimation of *resolution* to be expected. Finally, *product recovery* after elution classifies the chosen systems as suitable or not.

During an initial capture step, fast binding kinetics and high capacity for the desired product are important to ensure a high volumetric throughput and quantitative retention of the product molecule. During the following intermediate purification, sample throughput becomes less important in comparison to resolution. Both steps might thus be examined using finite bath studies. The final polishing step, however, has to assure high product purity and might thus require an optimization procedure based on packed column screening as outlined in the next section.

The general protocol for the establishment of isothermal or kinetic data is shown in Fig. 2. It consists of a matrix of process parameters comprising varying protein concentrations and buffer compositions (pH, ionic strength, salt type) incubated with the respective resins. The information is then derived from initial (C_0) and end (C_{eq}) concentrations of protein in the fluid phase. A defined setting of incubation

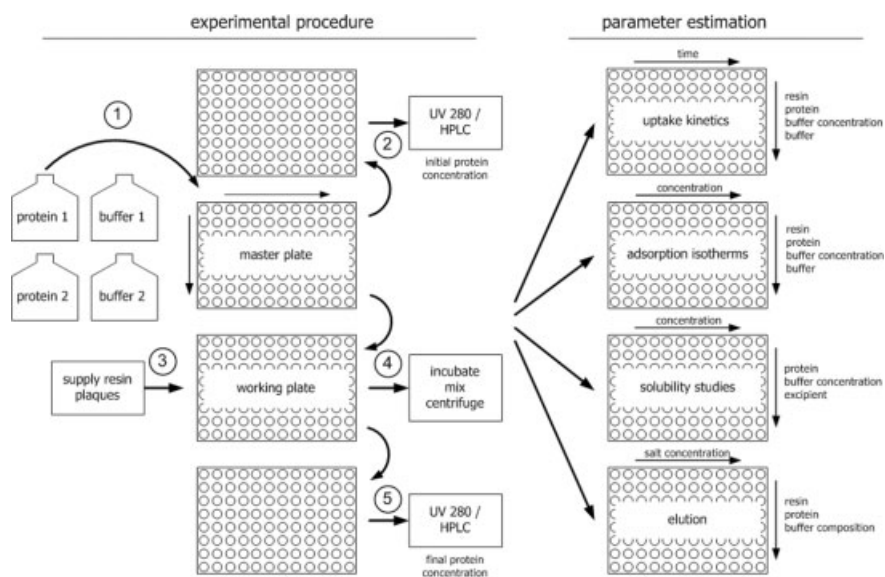


Figure 2. General HTS protocol: (1) protein solutions are prepared with changing concentrations in a master plate by aspirating and dispensing certain volumes from storage vessels of protein stock and buffer solutions; (2) samples from the master plate are analysed to determine initial protein concentration (optional); (3) resin plaques are supplied in a working plate and protein solution are transferred from the master plate; (4) incubation and mixing of the resin-liquid suspension, incubation process can either be stopped by centrifugation or vacuum; (5) samples of the supernatant are analyzed for determination of final protein concentration. When determining uptake kinetics protein solution from the master plate is transferred row by row at specific times, leading to a common endpoint for the multi-well plate. Solubility studies are performed without the need of adsorbent resins in the working plate.

times leads either to isothermal equilibrium data or kinetic uptake curves. An example of both data sets is given in Figs. 3a, b for isothermal capacity data and kinetic uptake of a monoclonal antibody on various commercial resins. Comparing the adsorbent types carrying the same ligand but exhibiting a different pore structure (SP Sepharose FF vs. SP Sepharose XL and Fractoprep SO_3 vs. Fractogel SO_3) clearly shows the high influence of adsorbent architecture. The latter is being neglected when restricting the screening approach to suitable ligands only as described in the first paragraph.

The evaluation of appropriate conditions for protein elution follows the same strategy by incubating pre-loaded adsorbents with varying buffer solutions (pH, ionic strength, salt type). Using increasing salt concentrations and thus applying micro-step wise elution allows the simulation of gradient elution conditions [19].

Finally, when designing a chromatographic process based on hydrophobic interactions, protein solubility under different salt conditions becomes a crucial parameter. By varying protein concentration vs. type and concentration of salt in the fluid phase one obtains a matrix indicating optimal conditions, which allow for protein adsorption under conditions ensuring protein solubility in the fluid phase. Fig. 4a shows the solubility of a monoclonal antibody as a function of antibody and salt concentration. The respective adsorption isotherms obtained for different hydrophobic resins are given in Figure 4b.

2.3 Column Screening

Resolution of chromatographic processes is closely connected to the number of theoretical plates present in a column or – more precisely – to the sequential resolution of a tremendous number of single adsorbent particles. Thus fine tuning of conditions expressing high resolutions are best carried out in column formats. The success of small-scale columns (1 mL – 5 mL) currently on the market results from their excellent applicability to chromatographic parameter screening and small scale purification tasks. Initially, kits containing a variety of resins packed in small-scale columns were marketed as a selection aid when determining optimal chromatographic parameter for a given purification task. They can easily be used with common chromatographic systems and offer a direct scalability to larger columns. A drawback, however, lies in the sequential screening format when compared to the highly parallelized resin screening performed on pipetting stations presented in the previous paragraph. However, in the era of combinatorial libraries, which led to a high number of small scale purifications these columns are now mainly used for the actual purification task rather than for pure process development. An overview summarizing the most common small scale screening kits is given in Tab. 1.

Generally, kits cover a specific chromatographic interaction mode – ion exchange or hydrophobic interaction – varying ligand type, density or matrix structure. Examples of this

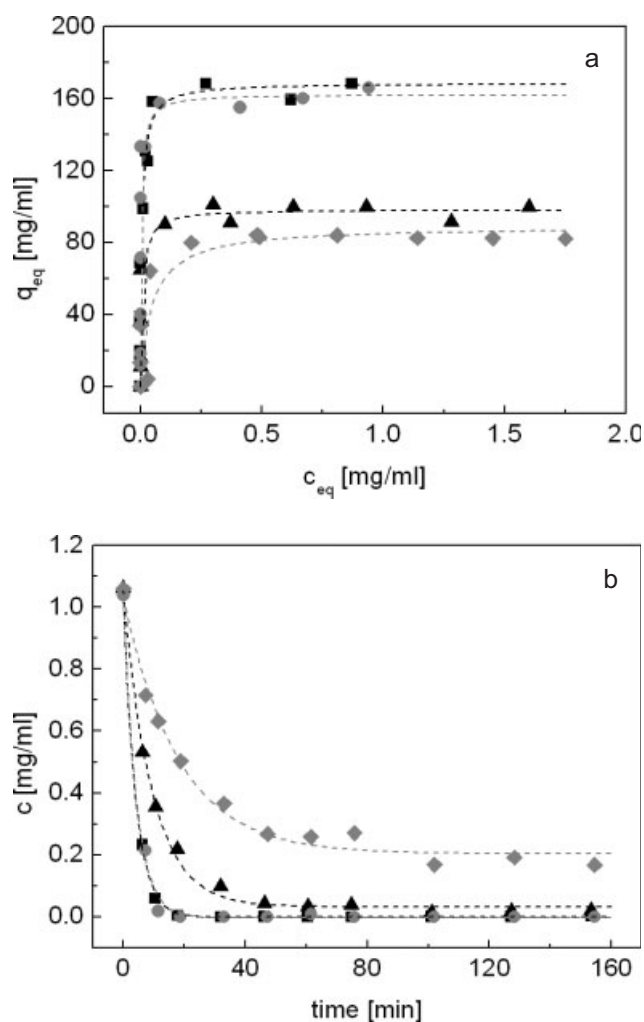


Figure 3. a) Adsorption isotherms of a monoclonal antibody on various CEX resins (◆ Fractoprep® SO₃⁻, ▲ SP Sepharose™ FF, ● Fractogel® EMD SO₃⁻, ■ SP Sepharose™ XL) in 50 mM MES (pH 6). The protein concentration on the stationary phase (q_{eq}) is plotted against the corresponding fluid phase concentration (c_{eq}). Data points are described through the Langmuir model (dotted line). b) Protein uptake kinetics of a monoclonal antibody on various CEX resins (◆ Fractoprep® SO₃⁻, ▲ SP Sepharose™ FF, ● Fractogel® EMD SO₃⁻, ■ SP Sepharose™ XL) in 50 mM MES (pH 6). The protein concentration (c) in the supernatant is plotted against time. Data points are described with an exponential decay (dotted line).

type of kit are, e.g., the commercially available HiTrap™ HIC Selection Kit (GE Healthcare) or ToyoScreen™ HIC Kit (Tosoh Bioscience), which both contain HIC resins expressing a range of different hydrophobicities. A typical screening result using the ToyoScreen™ HIC Kit is shown in Fig. 5.

The study shows the separation of ribonuclease A lysozyme and α -chymotrypsinogen using a set of six different HIC resins under identical fluid phase conditions. Evaluating the peak shapes, retention times and thus resolution the importance of resin screening for the development of chromatographic processes becomes obvious. The application of Butyl-650M facilitates almost a baseline separation whereas in the other extreme Hexyl-650C yields almost no resolution

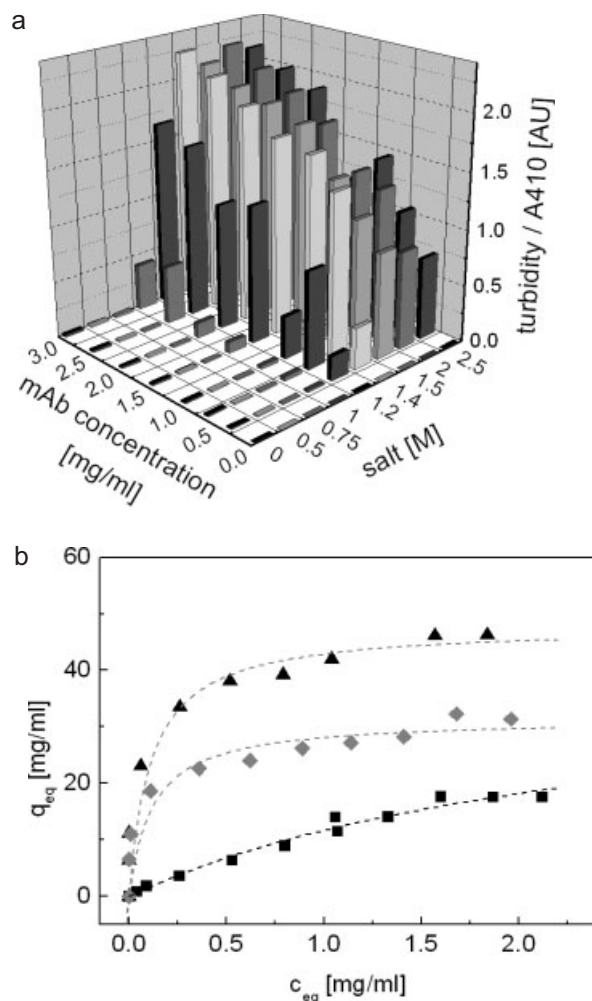


Figure 4. a) Protein solubility study: Precipitation of a monoclonal antibody is determined by turbidity measurements in 0.1 M Potassium Phosphate buffer (pH7) with varying (NH₄)₂SO₄ concentration. This antibody is only soluble in a buffer containing less than 1 M (NH₄)₂SO₄. b) Adsorption isotherms of a monoclonal antibody on various HIC resins (▲ Toyopearl® Hexyl-650C, ◆ Toyopearl® Butyl-650M, ■ Toyopearl® Phenyl-650M) in 0.1 M Potassium Phosphate buffer (pH7) plus 0.4 M (NH₄)₂SO₄. The protein concentration on the stationary phase (q_{eq}) is plotted against the corresponding fluid phase concentration (c_{eq}). Data points are described through the Langmuir model (dotted line).

and very flat peaks. Moreover, the study reveals that a simple ranking of hydrophobicities (proteins, resin, salt) might not be sufficient for determining parameters in HIC as the optimal conditions were found to lie centred within the range of resins examined.

The ToyoScreen™-system has further been used for the screening of chromatographic media examining the purification of a monoclonal Anti-TSH antibody [32]. The screening aimed for achieving a high peak resolution using four out of six HIC resins present in the ToyoScreen™ Kit: Toyopearl® Ether-650M, PPG-600M, Phenyl-650M and Butyl-650M. In this study, the phenyl-type resin showed the highest selectivity and was thus selected for optimization of mobile phase

Table 1. Overview of different resins used in commercially available screening kits from GE Healthcare and Tosoh Bioscience.

	Ligand	Ligand density (HIC)/Ion exchange capacity (IEX)	Dynamic binding capacity [mg/mL medium]	Bead size [μm]	Bead structure
HiTrap™ (GE Healthcare)					
<i>HIC Selection Kit</i>					
Phenyl sepharose High Performance	Phenyl	25 ^{A)}		24 – 44	6 % cross-linked agarose, spherical
Phenyl sepharose 6 Fast Flow (low sub)	Phenyl	20 ^{A)}		45 – 165	6 % cross-linked agarose, spherical
Phenyl sepharose 6 Fast Flow (high sub)	Phenyl	40 ^{A)}		45 – 165	6 % cross-linked agarose, spherical
Butyl Sepharose 4 Fast Flow	Butyl	50 ^{A)}		45 – 165	4 % cross-linked agarose, spherical
Octyl Sepharose 4 Fast Flow	Octyl	5 ^{A)}		45 – 165	4 % cross-linked agarose, spherical
<i>IEX Selection Kit</i>					
SP Sepharose Fast Flow	Sulfopropyl	0.18 – 0.25 ^{B)}	70 ¹⁾	45 – 165	6 % highly cross-linked agarose
SP Sepharose XL	Sulfopropyl	0.18 – 0.25 ^{B)}	> 160 ²⁾	45 – 165	6 % highly cross-linked agarose with bound dextran
CM Sepharose Fast Flow	Carboxymethyl	0.09 – 0.13 ^{B)}	50 ¹⁾	45 – 165	6 % highly cross-linked agarose
Q Sepharose Fast Flow	Trimethylamino	0.18 – 0.25 ^{C)}	120 ³⁾	45 – 165	6 % highly cross-linked agarose
Q Sepharose XL	Trimethylamino	0.18 – 0.26 ^{C)}	> 130 ⁴⁾	45 – 165	6 % highly cross-linked agarose with bound dextran
DEAE Sepharose Fast Flow	Diethylaminoethyl	0.11 – 0.16 ^{C)}	110 ³⁾	45 – 165	6 % highly cross-linked agarose
ANX Sepharose 4 Fast Flow (high sub)	Diethylaminoethyl	0.13 – 0.17 ^{C)}	43 ⁴⁾	45 – 165	4 % highly cross-linked agarose
ToyoScreen™ (Tosoh Bioscience)					
<i>HIC Type</i>					
Ether-650M	Ether		10 – 30 ²⁾	40 – 90	Methacrylate
PPG-600M	Polypropylen glycol		20 – 35 ²⁾	40 – 90	Methacrylate
Phenyl-650M	Phenyl		30 – 50 ²⁾	40 – 90	Methacrylate
Butyl-650M	Butyl		30 – 50 ²⁾	40 – 90	Methacrylate
SuperButyl-550C	Butyl		52 – 70 ⁵⁾	50 – 150	Methacrylate
Hexyl-650C	Hexyl		30 – 50 ²⁾	50 – 150	Methacrylate
<i>IEC Type</i>					
DEAE-650M	Diethylaminoethyl	0.08 - 0.12 ^{D)}	25 – 35 ⁴⁾	40 – 90	Methacrylate
SuperQ-650M	Quaternary amine	0.20 - 0.3 ^{D)}	105 – 155 ⁴⁾	40 – 90	Methacrylate
QAE-550C	Triethylaminoethyl	0.28 - 0.38 ^{D)}	60 – 80 ⁴⁾	50 – 150	Methacrylate
CM-650M	Carboxymethyl	0.08 - 0.12 ^{D)}	30 – 50 ²⁾	40 – 90	Methacrylate
SP-650M	Sulfopropyl	0.13 - 0.17 ^{D)}	40 – 60 ²⁾	40 – 90	Methacrylate
SP-550C	Sulfopropyl	0.14 - 0.18 ^{D)}	80 – 120 ²⁾	50 – 150	Methacrylate

Units: A) [$\mu\text{mol/mL gel}$], B) [$\text{mmol H}^+/\text{mL medium}$], C) [$\text{mmol Cl}^-/\text{mL medium}$], D) [eq/L_{gel}] – Densities/capacities measured with 1) Ribonuclease A; 2) Lysozyme; 3) Human serum albumine; 4) Bovine serum albumin; 5) γ -Globulin – Column dimensions for ToyoScreen™ columns: 1 mL columns: $0.64 \times 3 \text{ cm}$, 5 mL columns: $14.6 \times 3 \text{ cm}$; column dimensions for HiTrap™ columns: 1 mL columns: $0.7 \times 2.5 \text{ cm}$, 5 mL columns: $1.6 \times 2.5 \text{ cm}$

conditions. The conditions yielding qualitatively the best separation of antibody from cell culture contaminants were then used for the scale-up to a 10 mL semipreparative column yielding similar resolution results.

Although a selection of several media present in HiTrap™ Kits have been used in published work only few attempts were directed towards a comprehensive process development. Penna et al. [33] used the system for purification

of recombinant green fluorescent protein (GFPuv) following a three phase partitioning step. The resin screening included Butyl 4 FF, Octyl 4 FF and Phenyl 6 FF (HiTrap™ HIC Selection Kit) and a methyl type resin (Macro Prep HIC; Bio-Rad, Hercules, CA). As a result, the methyl type resin proved to be best suited for high purity leading to a recovery of the GFPuv in the HIC step of about 90 % and an 8-fold enrichment of antibody in relation to background proteins.

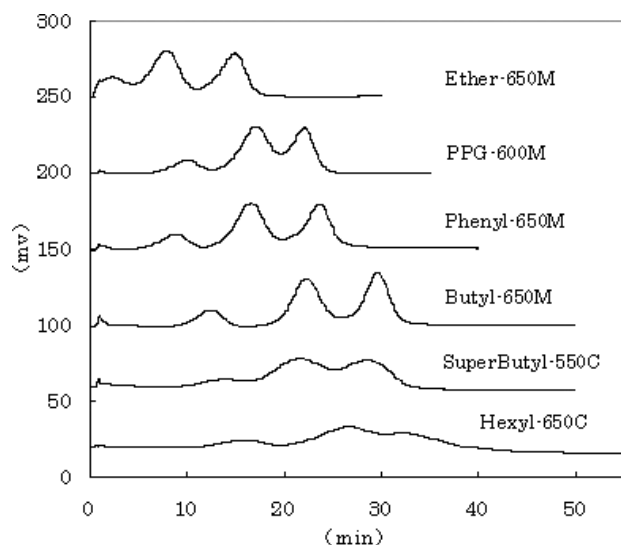


Figure 5. Screening of different Toyopearl® HIC resins: Separation of ribonuclease A, lysozyme and α-chymotrypsinogen; Column volume: 1 mL; Injection volume: 50 µL; Concentration (each protein): 1 mg/mL; Binding conditions: 0.1 M phosphate buffer + 1.8 M sodium sulphate at pH 7.0; Gradient length: 30 CV (data supplied by Dr. Egbert Müller, Tosoh Bioscience, Germany).

As a systematic comparison of small-scale selection kits still remains to be done – a first example is given in Fig. 6. Clearly, given the similar concept of column design among various manufacturers, small-scale screening columns can be expected to realistically simulate large scale column performance and are thus best suited for studies aiming for optimal resolution. A rational screening strategy might thus include a wide screening based on the high throughput approach for optimal solid-liquid phase combinations described in the previous section followed by fine tuning of suitable systems using small-scale columns packed with the respective adsorbent type. Following this approach screening could be performed with a low amount of expensive automation regarding the chromatographic system used.

In contrast to this approach I et al. [34] presented a study where automation was in the focus of the screening approach. The system aimed for a column based high throughput approach using preparative HPLC for protein separations. In this study an artificial intelligence “expert system” capable of planning, conducting and evaluating HPLC runs by varying solid and mobile phase parameters with the aim of optimizing protein resolution was implemented. One advantage of this procedure is the high degree of automation. There is thus only little need for human input which results in lower development costs and a higher machine utilization. The second advantage is the fact that the computer program uses real time analysis enabling an automatic and dynamic change of experimental parameters during the screening procedure. The major drawback of this technique lies in the pure restriction to column experiments yielding a rather low sample throughput.

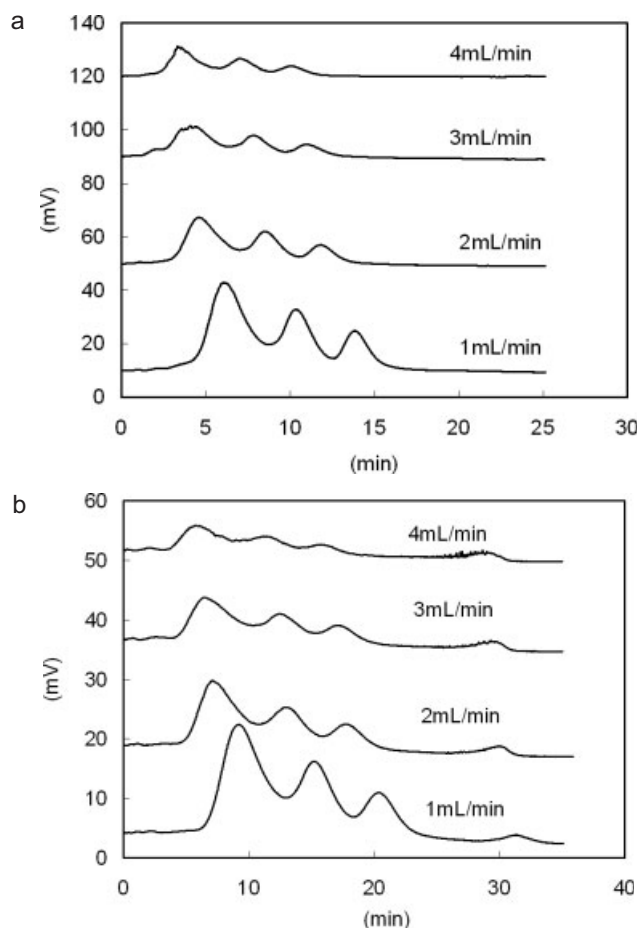


Figure 6. Effect of the flow rate on the separation of standard proteins; Column: a) ToyoScreen™ SuperQ-650M, b) HiTrap™ Q Sepharose; Column volume: 1 mL; Eluent A: 20 mM Tris-HCl (pH 8.0); Eluent B: 20mM Tris-HCl + 0.5 M NaCl (pH 8.0); Flow Rates: 1, 2, 3, 4 mL/min; Proteins: transferrin, ovalbumin, trypsin inhibitor; Protein concentration: 1mg/mL; Gradient: 30 min Linear; Injection Volume: 50 µL (data supplied by Dr. Egbert Müller, Tosoh Bioscience, Germany).

3 Analysis

The major bottleneck of most applications using HTS lies in the effectiveness of sample analysis and subsequent use of generated information, i.e., data analysis and design of further experiments.

A necessary prerequisite for the successful application of HTS techniques thus lies in the availability of analytical techniques which are not only selective and sensitive but furthermore fast, parallelizable, automatable and which require only small amounts of protein. In general, resolution and information content of an analytical technique are proportional to the time needed for analysis. One can further distinguish between analytical techniques, which can only be performed in sequential mode (e.g., column chromatography) and techniques offering parallel processing in a multi-well format (e.g., Enzyme Linked Immuno Sorbent Assay – ELISA). In most cases the parallelization of an ana-

lytical technique leads to a clear compensation of potentially higher required time per sample when compared to sequential techniques.

Due to the high number of parameters screened for in the first stage simple and fast assay techniques are the method of choice even though analytical sensitivity has to be compromised. Typical techniques for the determination of protein content are UV absorption or the use of dyes for protein staining such as the Bradford [35] or BCA assays. If the protein to be screened for is an enzyme, catalytic activity can be measured using labeled substrates or indirect sensor systems. A significant increase in the number of labels and chromophores has broadened the spectrum of detectable enzymes during the last few years [36]. The exclusion of unsuitable systems after the initial screening run offers the possibility to increase “time per sample” and thus analytical sensitivity. This concept is shown in Fig. 7.

Immunoassays are an analytical family, which combine high sensitivity, the use of crude samples and the possibility of parallelization [37]. A major advantage of immunoassays is that once the interacting antibody has been raised no *de novo* assay design is necessary for each new analytical task facilitating fast changes of the product to be screened for. A common technique within this family is the Enzyme Linked Immuno Sorbent Assay (ELISA), which can be automated using microtiter plates. The major drawbacks of microtiter plate ELISAs are lengthy periods of incubation [38] and loss of target protein using plate washers [39]. One solution to these problems is the use of membrane based ELISA protocols. The supernatant is withdrawn using a vacuum device. In this way the risk of losing target protein is minimized [39] but a typical 96-well plate assay still takes 3 hours [40]. Magnetic bead assays have been integrated into the major automation platforms such as Thermo KingFisher, Tecan Genesis, Tecan Freedom EVO and Beckman Biomek FX [41]. They offer the advantage of low background levels, large specific surfaces yielding large capacities and very short incubation times leading to a decreased total assay time of merely 45 minutes [38].

If the selectivity of an immunoassay alone is not sufficient, which may be the case if the protein mixture consists of several proteins with similar structural features, which are recognized by the antibody, immunoassays can be coupled with HPLC systems [42]. This procedure offers a higher resolution and a greater selectivity than standalone immunoassays but here again HPLC is the limiting step. Capillary electrophoresis (CE) has frequently been used for analysis of biological systems due to high speed of separation, high resolution and high mass sensitivity [43]. Coupling CE with immunoassays and laser-induced fluorescence detection, proteins can be specifically and rapidly detected [44–46].

As the actual reaction times of these techniques are set, new approaches aim at the miniaturization of the system minimizing sample load and thus “time per sample” required. Micro-total analysis systems (μ TAS, also known as Lab-on-a-chip) offer a further degree of automation. In this way, protein identification via MALDI-TOF [47], SDS-gels and even separation in 2 dimensions [48,49] could be achieved. Immunoassays have also been used in micro-scale. A microarray-based ELISA has been presented by Mendoza et al. [50].

4 Data Analysis and Experimental Design

Lab robotic platforms help to standardize and speed up screening experiments by automating most of the routine tasks, which are otherwise to be carried out manually. Commercial systems are supplied with software for the control and timing of previously specified robotic operations. User interfaces are typically provided for the input of elementary hardware actions, such as plate transfer, liquid pipetting, probe mixing and analysis. The experimental results are exportable in raw data formats.

However, the functionality of the available robotic software covers neither the rational design of experiments nor advanced data analysis. In more detail, the existing software must be extended to support data organization, parameter

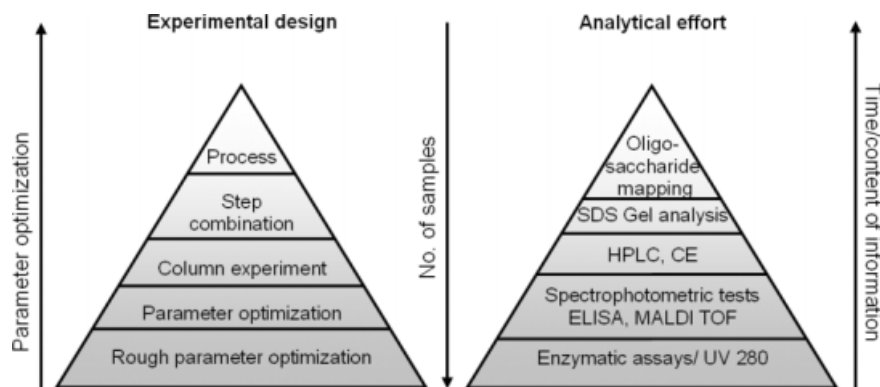


Figure 7. Analytical strategies in HTS: Correlation between number of samples to be handled, time and content of the information gathered via high throughput analytical techniques.

identification, error estimation, model discrimination and experimental design. In order to improve efficiency and to speed up the entire screening process, it is crucial to aim at the automation not only of manual operations but also of the brain work.

4.1 Data Organization

High throughput screening techniques – particularly if performed in parallel and on a daily basis – yield large amounts of raw data [51, 52]. This information has to be organized and efficiently stored in databases in order to be accessible for further analysis. Furthermore, each individual data set is to be supplemented by the corresponding experimental protocol with all relevant parameters for future assignment and consistent interpretation of deduced results. Data import and export functionality is typically required to cover various spreadsheet and plain text file formats.

4.2 Parameter Identification

Isolated experiments are often self-contained, in the sense that measured values are final results. In contrast, the outcome of serial experiments is typically subject to model based data analysis. The identification of unknown model parameters is certainly one of the most important and common techniques for advanced data analysis and evaluation. High throughput techniques are particularly suitable for the rapid acquisition of data sets, which are required for parameter identification of isotherms and kinetic expressions.

The identification process is based on systematic variation of the parameters, with the aim to minimize the difference between model predictions and measurement results, usually in the sense of least squares. Hence, identified parameters, for instance kinetic parameters, capture the essence of several measured values, for instance concentration values over time, which are determinable by independent experiments. This independency enables application of parallel high throughput screening formats, such as 96-well plates, which are superior to sequential methods.

4.3 Error Estimation

The estimation of errors, which are propagated from experimental data to derived quantities, is essential for the reliable interpretation of results, such as identified model parameters. Besides parameter identification, which involves an iterative optimization process, many results are directly deduced from measurement data by explicit mathematical formulae.

The Gaussian law is most commonly applied to the estimation of propagated standard deviations, although the results are inexact for non-linear models and for non-normally

distributed data, both of which is often the case. The same is true for the linear estimation of propagated covariances on base of the model's Jacobi matrix. Alternative statistical methods, such as Monte Carlo simulation and Bayesian inference, are less restricted and more flexible, however, for the price of increased computational effort [53, 54].

A principal component analysis of the error ellipsoid, which is a geometrical representation of the covariance matrix, provides information on which linear combinations of the model parameters are exceptionally well or poorly determined. Confidence intervals and correlations are, in addition to standard deviations and covariances, important measures of statistical data analysis. Finally, statistical tests, such as the χ^2 test, indicate whether or not the model is capable of describing the given data.

4.4 Experimental Design

The rational design of experiments must be an integral part of high throughput screening, since effectiveness of the experimental strategy is, besides speed and number of the individual experiments, most crucial for overall performance. However, intuitive design of high throughput experiments is limited by the combinatorial complexity of possible experiments, caused by the large number of parameters which are typically investigated at the same time.

Brute force concepts and factorial designs require a huge number of experiments in order to screen the entire parameter space and are, hence, not very effective. Partial factorial designs reduce the number of experiments; however, the gained information is often suboptimal. Consequently, optimal experimental design aims at maximizing the information, which is gained from a predefined number of experiments [55, 56]. A priori information, typically a model with initial parameter values, is analyzed in order to determine a set of individual experiments, which are optimal with respect to a predefined criterion.

Consider, for example, the refinement of model parameters, which are identified from experimental data: Initial parameter values are obtained either form an educated guess or from preliminary experiments. Further experiments are then designed such as to enable most accurate parameter identification, where accuracy is quantified by specific measures of the expected error ellipsoid, such as volume or largest principal component [57].

Maximization of accuracy is, however, not always a wise target. In order to optimize utilization of experimental resources, it is often more advantageous to design as few experiments as possible such as to achieve a desired accuracy. Here, the actual results need to be carefully checked for the specified accuracy, since experimental errors may always interfere with the expected outcome [57].

Furthermore, the choice of initial values significantly impacts on the optimal design. This influence is reduced by robust experimental design, however, for the price of non-

optimality with respect to the original design criterion. Another approach is to implement optimal experimental design as an iterative procedure, which benefits from considering the result of each iteration when designing the next (see Fig. 8). Here, successive designs are no longer independent and parallelization is limited by the number of individual experiments per iteration.

Parallelization limits also apply to classical optimization schemes, such as simplex or genetic algorithms [58]. These algorithms are typically chosen for process optimization, when appropriate models are not available. Consider, for instance, a chromatographic separation process, which is characterized by adsorbent type and design, buffer composition, pH, ionic strength and salt types, amongst various other process parameters. Derivative based algorithms, for example the gradient method, are often not suitable for bioprocess development, since the available data are too noisy for numerical derivation. Conversely, parameter sensitivities, when available, provide valuable information for rational process optimization [59].

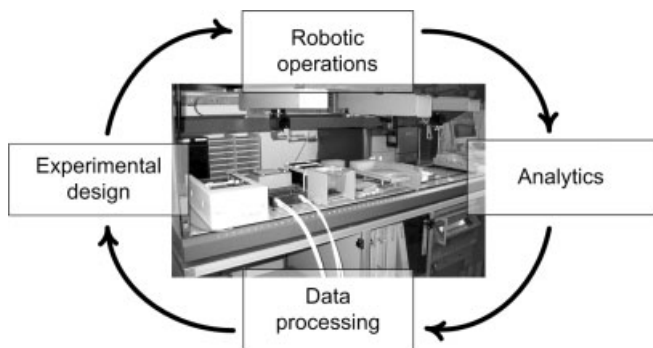


Figure 8. Iterative circle of high throughput process development.

4.5 Model Discrimination

The exact mechanisms of observed phenomena are often not known a priori and remain, hence, to be experimentally determined. Moreover, several mechanistic models, which vary in their depth of detail, are typically available for the description of one and the same physical phenomenon. The variety of candidate models, which are available for model based data analysis and interpretation, becomes even more diverse, when phenomenological models are considered.

The framework of model discrimination enables identification of individual models, which are most suitable for the analysis of specified experiments [55, 56]. In the context of high throughput screening various candidate models occur, for example, in terms of different isotherms, such as Freundlich and Langmuir.

In practice, candidate models tend to behave very similar through most of the experimental parameter space. Hence, model discrimination aims at designing experiments with maximal expected difference of the model predictions,

weighted by the expected confidence of the corresponding experimental results [60]. Here, the iterative approach to optimal experimental design is particularly advantageous, since slight differences between the predictions are potentially blurred by inevitable inaccuracies of the initial values.

5 Conclusions

Modern process development is highly determined by stringent “time to market” restrictions and thus the need for fast but at the same time optimal process development. The increasing complexity of processes triggered partly by the introduction of new materials leads, however, to a multi-dimensional parameter space, which is difficult to explore by conventional techniques.

This problem can be successfully tackled by the use of HTS technology for process development in chromatography. The successful approach when implementing HTS screening procedures for chromatographic parameter estimation points towards a more general application of HTS techniques to other unit operations in downstream processing such as membrane separations, aqueous two phase systems or crystallization.

The success of this approach is, however, highly dependent on the realization of experimental techniques expressing a high degree of parallelization and automation suitable for HTS, the respective tools for sample analysis and data handling and finally powerful strategies for experimental design as a fundamental platform.

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